European Network of
Transmission System Operators
For Electricity

Upgrade Guide

Energy Communication Platform
Version 4.16.0

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Energy Communication Platform, version 4.16.0

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2. Executive Summary

This document provides upgrade instructions for all ECP components.

This document is primarily targeted at Administrators.

3. Revision History

Version	Date	Description
4.1.3	20.11.2017	Initial version
4.1.4	05.02.2018	Documentation review
4.2.0	21.03.2018	Version update
4.3.0	25.06.2018	Version update
4.3.0	24.09.2018	CD/endpoint uninstallation via admin console
4.3.0	12.11.2018	Data directories paths update
4.4.0	13.06.2019	Updated for the ECP 4.4.0 version
4.5.0	27.08.2019	Version update
4.6.0	24.10.2019	Version update
4.6.1	18.03.2020	Version update
4.7.0	28.06.2020	Upgrade procedure changes
4.7.1.	31.07.2020	Version update
4.7.2	16.11.2020	Docker upgrade changes and ECP Broker upgrade changes
4.8.0	23.03.2021	Java 11 upgrade changes
4.8.1	09.07.2021	Version update
4.8.2	16.12.2021	Windows Server 2019, supported browsers update, ActiveMQ version update, Internal brokers configuration, optional Java 11 installation, formatting, JMS directory parameter
4.9.0	26.05.2022	Supported browsers update, ActiveMQ version update, Java version update, AMQPS configuration
4.10.0	07.11.2022	Hibernate, Broker ioExceptionHandler, Update configuration changes, CentOS 8 deprecation, Java 11.0.16 disclaimer

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4.11.0	16.05.2023	Update configuration changes, Keycloak and LDAP in WS authentication, update of Upgrade Notes and GUI chapters, Java 17, update of supported OSs and DBs, added chapter New configuration properties, Kubernetes
4.12.0	27.10.2023	Update configuration changes, New Prometheus metrics, message priority configuration
4.14.0	10.06.2024	Artemis, CSRF secret property, CD and Broker Windows removal, Keycloak support removal
4.15.0	21.10.2024	Windows Server 2022 addition, Windows Server 2016 removal, Internal Broker upgrade, ECP Broker Upgrade, ECP Endpoint start-up time note
4.16.0	06.05.2025	Log timestamps pattern change, LDAP server side authentication

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4. Service Level Requirements

Please note that the following HW requirements is only estimation, and they may be changed according to performance requirements. In general, we recommend the usage of virtual machines, which can be resized when more resource is necessary.

Real HW requirements also depend on message traffic.

4.1. Recommended Minimum Hardware Requirements

4.1.1. Endpoint with Internal Broker

For an Endpoint, which will at peak send a few thousands of messages per hour, we recommend the following configuration.

CPU	4 cores – each at least at 2Ghz
RAM	8 GB
HDD	For ECP: 40 GB, For Message Archive: depends on the message traffic

4.1.2. Broker

For a Broker, which will exchange messages between 50 Endpoints, each sending at peak few thousands of messages per hour, we recommend the following configuration.

CPU	8 cores – each at least at 2Ghz
RAM	16 GB
HDD	100 GB

4.1.3. Component Directory

To support network of up to 50 endpoints, we recommend following hardware configuration for Component Directory.

CPU	2 cores – each at least at 2Ghz
RAM	4 GB
HDD	10 GB

4.2. Supported Operating Systems

- *Windows Server 2019*
- *Windows Server 2022*
- *RHEL 8*
- *RHEL 9*
- *Oracle Linux 8*
- *Oracle Linux 9*

4.3. Supported Java Version

- *Java 17 is required. Recommended Java provider is Red Hat.*
- *Linux:*
  - *Using Unix repositories – yum install java-17-openjdk*
- *Windows:*
  - *When OpenJDK JRE package is used, please make sure that only JRE_HOME system variable is defined.*

4.4. Supported Web Browsers

Google Chrome 133.0.6943.142 (64-bit), Firefox 135.0.1, Safari 18.3, MS Edge 133.0.3065.92

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4.5. Supported Resolution

Minimal supported resolution is 1280x1024.

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5. ECP Endpoint Upgrade

If you are upgrading Endpoint in the High Availability installation, ECP does not support an upgrade with a zero downtime – it is not possible for the running nodes to have a different ECP versions. This procedure is recommended for an upgrade with a minimal outage:

1. *Stop one of the Endpoint nodes*
2. *Upgrade the stopped node*
3. *Stop all the other nodes*
4. *Start the upgraded node*
5. *Upgrade all remaining nodes and start them*

The upgrade process of ECP Endpoint which still uses an embedded ActiveMQ Internal Broker (ECP Endpoint 4.12 or lower) does not move messages from the ActiveMQ to the new Artemis Internal Broker. Therefore, all pending messages would be lost after the upgrade.

To prevent the possible data loss, it is strongly recommended to disconnect the ECP Endpoint from ECP Broker and Business Application before the upgrade and proceed with the upgrade only when all internal queues are empty.

5.1. Upgrade on Windows

5.1.1. Variables used in the guide

Variables used throughout this upgrade guide are using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<install_path>	Path pointing to the installation directory

5.1.2. Stop application

5.1.2.1. Using Command Line

a) If installed as a service:

> *Stop service using*

- *SC stop ecp-endpoint*

b) If not installed as a service:

> *Stop application using*

- *<install_path>\tomcat\bin\shutdown.bat*

5.1.2.2. Using GUI

a) If installed as a service:

> *Start > Run > "services.msc"*

> *Find service with name like "ecp-endpoint" and description "ECP endpoint"*

> *Stop service by button on the top left*

b) If not installed as a service:

> *Stop application by using*

- <install_path>\tomcat\bin\shutdown.bat

5.1.3. Java

Check the current Java version using the following command:

> ☐ *java -version*

The expected output is *openjdk version "17.0.14"* or newer versions. If the version is below *17.0.14* or the 0.14-update is older than 6 months, the update is required.

5.1.3.1. Upgrade to Java 17 (only when upgrading from ECP 4.10.1 or lower)

In case EDX is deployed on the same server, perform EDX upgrade in parallel with ECP upgrade – both applications are using same Java.

Current installed Java 11 has to be uninstalled before the installation of Java 17. Navigate to Windows "Programs and Features", select Java 11 and "Uninstall".

Update of Java is necessary prior installation of new application version. If you do this in the wrong order, your application might need to be reinstalled. Newer version of Java OpenJDK JRE can be obtained from official Red Hat webpages on the following address - https://developers.redhat.com/products/openjdk/download

During the installation, the following options must be selected:

> ☐ *Windows Registry*

> ☐ *PATH Variable*

> ☐ *JAR Files Association*

The following options must not be selected:

> ☐ *JAVA_HOME Variable*

> ☐ *REDHAT_JAVA_HOME Variable*

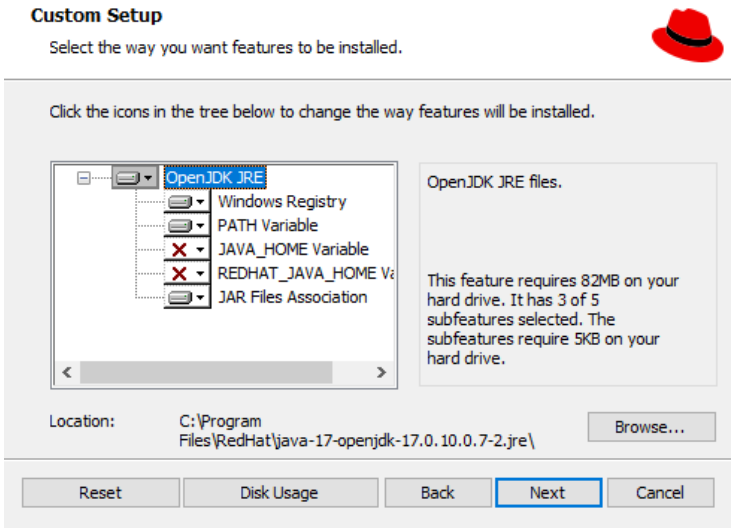

Figure 1: OpenJDK Installation

After Java installation, configure Java properties:

> ☐ *Open the security configuration file, e.g. on path: C:\Program Files\RedHat\java-17-openjdk-17.0.14.0.7-1.jre\conf\security\java.security*

> ☐ *Change the line securerandom.source=file:/dev/random to securerandom.source=file:/dev/urandom*

> ☐ *Save the file*

CONFIGURE JRE_HOME

> ☐ *Open explorer*

> ☐ *Right click on This PC and select Properties*

> ☐ *Select Advanced system settings -> Environment Variables -> System Variables -> New…*

- *Name: JRE_HOME*

- *Value: path to installed Java (e.g. C:\Program Files\Java\RedHat\ java-17-openjdk-17.0.14.0.7-1.jre\)*

CHECK JAVA_HOME IN NOT SET

> ☐ *Open explorer*

> ☐ *Right click on This PC and select Properties*

> ☐ *Select Advanced system settings -> Environment Variables -> System Variables*

> ☐ *Make sure that the JAVA_HOME system variable is not configured*

- *If JAVA_HOME system variable is set, remove it using the Delete button*

5.1.4. Install new version of application

For installation of the new version .jar go through following steps.

The database scheme update is done automatically after the start of the application.

Before upgrading the application, please make sure that only JRE_HOME system variable is defined, if OpenJDK JRE package is used.

GUI INSTALLATION

1. ☐ *Double-click the <ecp-jar> file. If Java was registered to execute jar files, an installer window should open. Skip to step 5. If an installer window does not open, follow steps 2-4:*
2. ☐ *Open command line with the administrator rights:*

   > *Start > type "cmd" > right click > run as administrator*

3. ☐ *Change directory to location of **ecp-endpoint.jar**, e.g.*

   > *cd c:\ecp-installer*

4. ☐ *Run the **ecp-endpoint.jar** file using*

   > *java -jar <ecp-jar>*

5. *ECP Endpoint installer window is opened.*

   > ☐ *Select installation type – Upgrade an existing ECP installation*

   > ☐ *Select target installation path (the target directory must exist)*

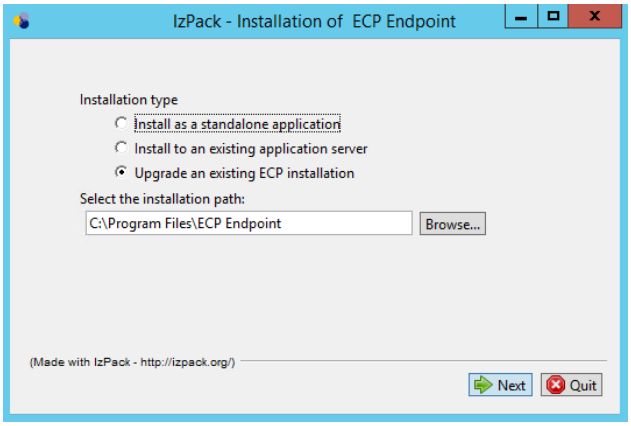

Figure 2: Installation 1

> ☐ *Click Next*

> ☐ *Installation is processed*

> ☐ *Click Next*

> ☐ *If desired, an automatic installation script can be generated. It can be used in the future to install the application using the same options.*

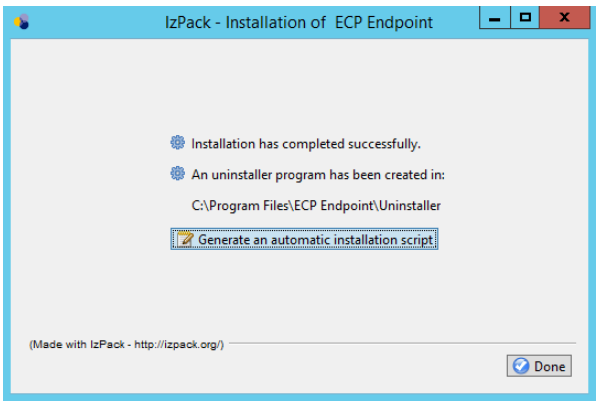

Figure 3: Installation 2

> ☐ *Click Done*

COMMAND LINE INSTALLATION

> ☐ *Open command line with the administrator rights:*

- *Start > type "cmd" > right click > run as administrator*

> ☐ *Change directory to location of <ecp-jar>, e.g.*

- *cd c:\ecp-installer*

> ☐ *Run the ecp-endpoint.jar file using*

- *java -jar <ecp-jar> -console*

> ☐ *Follow the console installation wizard*

5.1.5. Internal Broker (when upgrading from ECP 4.12.0 or lower)

ECP Endpoint no longer contains an embedded ActiveMQ Internal Broker. Now an externally running Artemis Internal Broker is required. Artemis Internal Broker is provided as a separate component, with its own installer.

Please refer to the ECP Installation Guide, chapter *Internal Broker Installation* on how to install the Internal Broker. It is also possible to connect multiple ECP Endpoints to one Internal Broker, so you may use an already installed Internal Broker (see the *ECP Endpoint configuration* subchapter below on how to configure multiple ECP Endpoints).

If the ECP Endpoint uses the High Availability deployment, Internal Broker should be also installed in the High Availability deployment.

5.1.5.1. Internal Broker Artemis configuration

ActiveMQ broker configuration in *activemq-*.xml* files located in *<install_path>/jms* is no longer used. Instead, the Artemis Internal Broker uses the configuration in *broker.xml* file located in the Internal Broker installation:

> *<Internal Broker installation path>/broker/etc*

When custom changes were previously made in *activemq-*.xml*, it is necessary to migrate these changes to *broker.xml*. Artemis Internal Broker do not use a separate file for the direct messaging configuration (previously in the *activemq*-external*.xml file), everything is configured in one *broker.xml* file.

There are multiple *activemq-*.xml* files in ECP Endpoint, so for the migration use only the files which were used by ECP Endpoint.

> */jms/internal-ssl-jks/ if the optional Hardware Security Module (HSM) is not used and internalBroker.useAuthentication is set to true. This was a default configuration.*

> */jms/internal-ssl-hsm/ and /jms/external-ssl-hsm/ if HSM is used*

> */jms/ if optional HSM is not used and internalBroker.useAuthentication is set to false*

File was then selected to match the configured ECP application profile:

> *activemq-nonha(-external).xml when ecp-nonha profile is used. This was a default configuration.*

> *activemq-ha(-external).xml when ecp-ha profile is used*

> *activemq-inmemory(-external).xml when ecp-inmemory profile is used*

In a default configuration, the following file was used:

> */jms/internal-ssl-jks/activemq-nonha.xml*

Compare the matching elements in *activemq-*.xml* and *broker.xml*, and make changes in *broker.xml* if applicable:

> ***transportConnector*** *element in activemq-*.xml matches* ***acceptor*** *element in broker.xml*
>
> - *In the connection string, compare a port number and connection parameters (for example, TLS protocol and ciphers)*
> - *When direct messaging was used, create a new acceptor element with a different name and port number. This will be used for the direct messaging connections.*
>
> ***memoryUsage*** *element in activemq-*.xml matches* ***global-max-size*** *element in broker.xml*
>
> ***storeUsage*** *element in activemq-*.xml matches* ***max-disk-usage*** *element in broker.xml*
>
> ***destinationPolicy*** *element in activemq-*.xml matches* ***address-settings*** *element in broker.xml*

ECP Endpoint contains example ***broker.xml*** files in *<install_path>/jms/ssl-hsm* and *ssl-jks* folders. These files are not used by ECP Endpoint, they are meant only as a support for the Internal Broker ***broker.xml*** configuration. For example, they show how to configure the direct messaging, or queue memory limits.

If you require more details about the migration process and ***broker.xml*** configuration, please refer to the official Artemis Migration Guide and Configuration Reference.

Once the configuration migration is complete, all *activemq-*.xml* files can be removed.

5.1.5.2. ECP Endpoint configuration

Please refer to the ECP Installation Guide, chapter *Configuration of ecp.properties - Internal Broker configuration* on how to configure ECP Endpoint to be able to connect to Internal Broker.

5.1.6. Update configuration

5.1.6.1. When upgrading from ECP 4.12.0 or lower

When upgrading from ECP 4.12.0 or lower, the following changes are required.

> ☐ *Open <install_path>**ecp.properties**:*
>
> - *Add this property:*
>
> ▪ *ecp.csrf.secret*

- *Its value is used as a secret used in a creation of the CSRF token hash for GUI. It should contain random characters, like a password.*
- *These properties are no longer required, so they can be removed:*
  - *spring.activemq.user*
  - *spring.activemq.password*

> ☐ *Open <install_path>\tomcat\conf**server.xml**, and find HTTPS connector element*

> ☐ *Add the new HTTPS connector element:*

```
<Connector port="8443" protocol="eu.entsoe.ecp.connector.protocol.EcpHttp11NioProtocol"
sslImplementationName="org.apache.tomcat.util.net.jsse.JSSEImplementation"
maxThreads="150" SSLEnabled="true" maxParameterCount="1000" >
    <UpgradeProtocol className="org.apache.coyote.http2.Http2Protocol" />

    <SSLHostConfig protocols="TLSv1.3"
ciphers="TLS_AES_256_GCM_SHA384,TLS_CHACHA20_POLY1305_SHA256,TLS_AES_128_GCM_SHA256" >
        <Certificate certificateKeystoreFile="<path to keystore>"
certificateKeystorePassword="<keystore password>" type="RSA" />
    </SSLHostConfig>
</Connector>
```

> ☐ *Copy a value from the old HTTPS connector attribute and pass it to a matching attribute in the new HTTPS connector:*
  - *keystoreFile to certificateKeystoreFile*
  - *keystorePass to certificateKeystorePassword*
  - *any other attribute where a custom change was previously made (for example ciphers)*

> ☐ *Remove the old HTTPS connector element*

Keycloak authentication is no longer supported, so the Keycloak properties can be removed when the Keycloak authentication was previously enabled:

> *Open <install_path>**ecp.properties**:*
  - *Remove all properties starting with "keycloak."*
  - *Remove "ecp-sso" profile from the spring.profiles.active*

Please see the *How to configure Keycloak integration with LDAP* chapter in the *ECP Administration Guide* to make a switch to the LDAP authentication.

Optionally, you can remove the *-Decp.endpoint.jms.directory* property from the following files:

> ☐ *<install_path>\tomcat\bin**ecp-install-service.cmd***

> ☐ *<install_path>\tomcat\bin**ecp-updatel-service.cmd***

> ☐ *<install_path>\tomcat\bin**setenv.bat***

In case ECP is installed as s service, it is necessary to also execute the **ecp-udate-service.cmd**.

As of the Prometheus metrics, the list of previously used ActiveMQ metrics with their Artemis equivalent can be found in the *Artemis Prometheus metrics* chapter.

When Artemis Internal Broker does not use a default keystore, its public certificate must be imported to the ECP Endpoint keystore referenced in the *internalBroker.keystore.location* property. You can find a location of the Artemis keystore in the *keyStorePath* attribute of the acceptor URI in the *broker.xml* file. By default, it is in *<Artemis installation path>/broker/etc/keystore.jks*.

Alternatively, it is possible to configure Artemis Internal Broker to use the ECP Endpoint keystore from the *internalBroker.keystore.location* property instead.

5.1.6.2. When upgrading from ECP 4.11.0 or lower

When upgrading from ECP 4.11.0 or lower, the following changes are required.

> ☐ *To enable health endpoint metrics, add these lines in <install_path>**ecp.properties**:*

- *To enable the metrics:*
  - *management.endpoint.health.show-details=ALWAYS*

5.1.6.3. When upgrading from ECP 4.10.1 or lower

When upgrading from ECP 4.10.1 or lower, the following changes are required.

> *<u>When the LDAP authentication is enabled</u>*

- *The Web Services requests are no longer authenticated by using the users from the **users.properties** file.*
- ☐ *It is necessary to create new users in the LDAP server, or to change the Web Services requests to use users which already exists in the LDAP server.*

5.1.6.4. When upgrading from ECP 4.9.0 or lower

When upgrading from ECP 4.9.0 or lower, the following changes are required.

Remove the *file:* prefix from *-Dspring.config.additional-location* property in the following files:

> ☐ *<install_path>\tomcat\bin**ecp-install-service.cmd***

> ☐ *<install_path>\tomcat\bin**ecp-update-service.cmd***

> ☐ *<install_path>\tomcat\bin**setenv.bat***

The property should then look like this:

> *-Dspring.config.additional-location=<install_path>\ecp.properties,<install_path>\ecp-users.properties*

In case ECP is installed as a service, it is necessary to also execute the ***ecp-update-service.cmd*** script.

5.1.6.5. MS SQL database

MS SQL database connector is now forcing TLS encryption by default. If you do not use TLS encryption, add *encrypt=false* parameter to the connection string in *ecp.db.url.*

If you use Windows authentication for MS SQL database, download Microsoft JDBC Driver 12.8.1 for SQL Server from https://repo1.maven.org/maven2/com/microsoft/sqlserver/mssql-jdbc_auth/12.8.1.x64/mssql-jdbc_auth-12.8.1.x64.dll. Copy the file to this directory and remove the older version of **.dll** file:

- *<install_path>\tomcat\bin*

5.1.7. Start application

5.1.7.1. Using Command Line

a) If installed as a service:

> *Start service using*

  – *SC start ecp-endpoint*

b) If not installed as a service:

> *Start application using*

  – *<install_path>\tomcat\bin\startup.bat*

5.1.7.2. Using GUI

a) If installed as a service:

> *Start > Run > “services.msc”*

> *Find service with name like "ecp-endpoint" and description "ECP Endpoint"*

> *Start service by button on the top left*

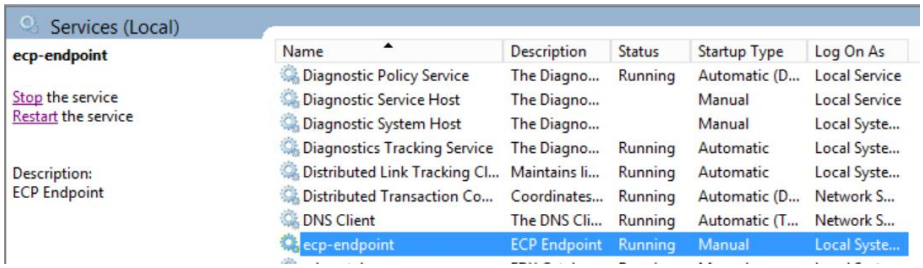

Figure 4: Services

b) If not installed as a service:

> *Start application by using*

- <install_path>\tomcat\bin\startup.bat

5.1.8. Installation verification

The application should be installed in the <install_path> folder. This installation folder should contain configuration files, Tomcat folder, and uninstaller. After the application is successfully started, it creates additional folders for data and log files. Default names and locations of these folders are *<install_path>\data* and *<install_path>\logs*.

5.1.8.1. Service status

The status of the service, if installed, can be checked via command line:

> *SC query ecp-endpoint*

or in the Windows Services tool.

5.1.8.2. Log files

Check the application log files for more information about its status. The default location is *<install_path>\logs*.

5.1.8.3. GUI

When all previous steps were successfully done it should be possible to open GUI in the web browser. The web browser refresh (Ctrl+Shift+R) or clearing of the web browser cache should be made to make sure that the new version features and changes are loaded.

Everything should be working and configured as it was before the upgrade (except the new version features and changes). Dashboard should be opened after logging into application.

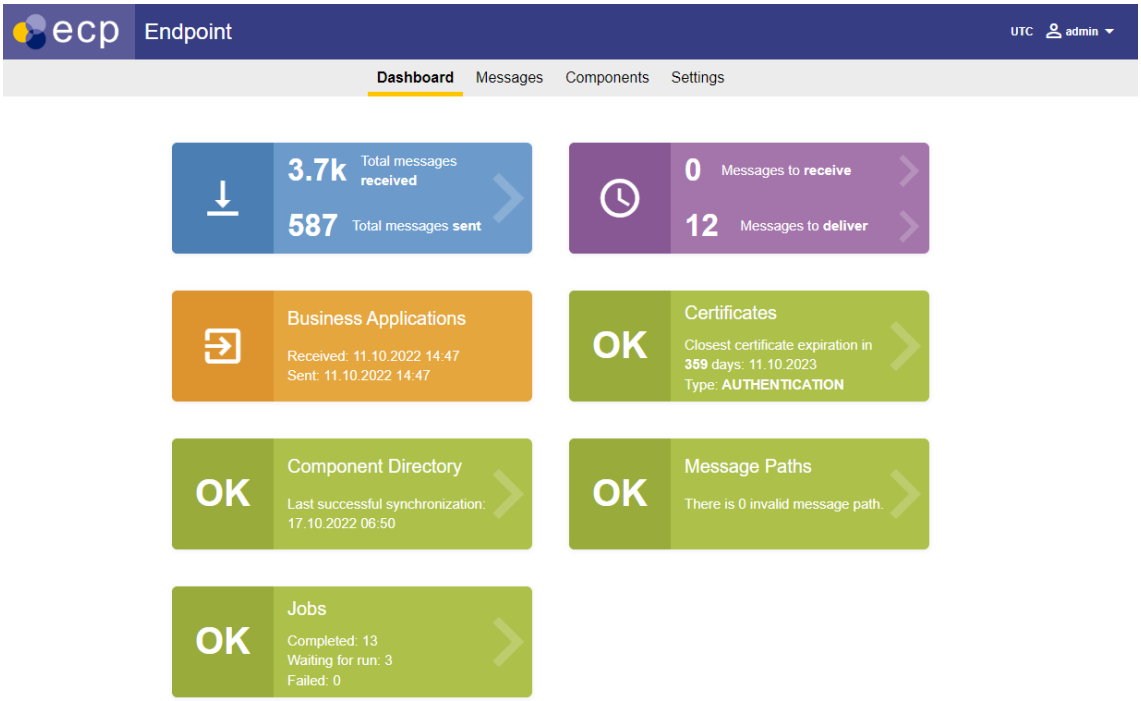

Figure 5: GUI - Verification

5.1.8.4. Apply functional smoke test if needed

Please refer to the *ECP Installation Guide*, chapter *ECP Endpoint Registration and Functional Smoke Test* to apply the test.

5.1.9. Upgrade Notes

Log record time zone is set to UTC, therefore there will be overlap of times in log files.

Default GUI time zone is changed to UTC. Times of old messages are not changed to the new time zone, therefore there can be overlap of times of old and new messages. Time zone of GUI can be changed from GUI Settings page.

All properties files, ***ecp-logback.xml***, ***authKeystore.jks*** and ***server.xml*** are never overwritten during the upgrade. Chapter *Update configuration* describes all necessary changes which must be made manually to keep these files up to date with the latest ECP version.

Due to a change of the messages indexes in the database, the first start-up after the upgrade may take a longer time when upgrading from ECP 4.14.0 or lower versions. It should take between 20 seconds to 2 minutes, but it depends on number of messages in the database. Reducing the number of messages in the database before the upgrade will help to reduce the start-up time.

ActiveMQ database tables are no longer used by ECP Endpoint, so these tables can be removed from the database:

- *ACTIVEMQ_MSGS*
- *ACTIVEMQ_ACKS*
- *ACTIVEMQ_LOCK*

The pattern of timestamps in the logs has been changed to comply with ISO8601. An example of the current pattern of a timestamp:

- *2024-05-20T14:30Z*

When LDAP authentication is used, the default authentication mode has been changed from the password comparison authentication to the bind authentication. Therefore, it is necessary to verify that the LDAP server supports this authentication mode. There is a new *ecp.ldap.useServerAuthentication* property to configure the authentication mode. This change does not affect the Active Directory authentication.

5.2. Upgrade on Linux

5.2.1. Variables used in the guide

Variables used throughout this upgrade guide are using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<install_path>	Path pointing to the installation directory

5.2.2. Stop application

The following command is available:

- *systemctl stop ecp-endpoint.service*

If not installed as a service:

- Stop application using bash script *"<install_path>/tomcat/bin/shutdown.sh"*

5.2.3. Java

Check the current Java version using the following command:

> ☐ *java -version*

The expected output is *openjdk "17.0.14"* or newer versions. If the version is below *17.0.14* or the 0.14-update is older than 6 months, the update is required.

5.2.3.1. Upgrade to Java 17 (only when upgrading from ECP 4.10.1 or lower)

In case EDX is deployed on the same server, perform EDX upgrade in parallel with ECP upgrade – both applications are using same Java.

An upgrade of Java is necessary prior installation of the new application version.

If your Java version is 11, run the following command to install Java 17:

> ☐ *yum install java-17-openjdk*

The minimal Java version is 17.0.14. If your Java version is lower, run the following command for Java update:

> ☐ *yum update java-17-openjdk*

After Java installation, configure Java properties:

> ☐ *Open the security configuration file, e.g. on path: /usr/lib/jvm/ java-17-openjdk-17.0.14.0.7-2.0.1.el9.x86_64/conf/security/java.security*

> ☐ *Change the line securerandom.source=file:/dev/random to securerandom.source=file:/dev/urandom*

> ☐ *Save the file*

! Do not uninstall Java 11 at this moment, First upgrade ECP Endpoint – follow the steps below. !

5.2.4. ECP Endpoint application upgrade

Change the directory to the folder where the ECP rpm package is located and then invoke rpm upgrade command.

> ☐ *rpm -U <ecp-rpm>*

It keeps original configuration files and it creates new configuration files with a ***.rpmnew*** suffix. File ***.rpmnew*** is not created in case that original configuration was not modified.

By default, ECP Endpoint is set up to automatically start after server start. The automatic start can be disabled by the following command:

> ☐ *systemctl disable ecp-endpoint*

Following folders are updated:

Folder	Description
/usr/share/ecp-endpoint	Application binaries folder. This is the default location of <install_path>
/var/lib/ecp-endpoint	Application data folder
/var/log/ecp-endpoint	Application log folder
/var/log/ecp-directory	Application log folder
/etc/ecp-endpoint	ECP configuration files (**ecp.properties**)
/etc/systemd/system/ecp-endpoint.service.d/	Default configuration of ECP endpoint. Used to set JVM configuration and location of configuration files (**env.conf**)

5.2.5. Internal Broker (when upgrading from ECP 4.12.0 or lower)

ECP Endpoint no longer contains an embedded ActiveMQ Internal Broker. Now an externally running Artemis Internal Broker is required. Artemis Internal Broker is provided as a separate component, with its own installer.

Please refer to the ECP Installation Guide, chapter *Internal Broker Installation* on how to install the Internal Broker. It is also possible to connect multiple ECP Endpoints to one Internal Broker, so you may use an already installed Internal Broker (see the *ECP Endpoint configuration* subchapter below on how to configure multiple ECP Endpoints).

If the ECP Endpoint uses the High Availability deployment, Internal Broker should be also installed in the High Availability deployment.

5.2.5.1. Internal Broker Artemis configuration

ActiveMQ broker configuration in *activemq-*.xml* files located in */etc/ecp-endpoint/jms* is no longer used. Instead, the Artemis Internal Broker uses the configuration in *broker.xml* file located in the Internal Broker installation:

> */opt/eccosp-artemis/etc*

When custom changes were previously made in *activemq-*.xml*, it is necessary to migrate these changes to *broker.xml*. Artemis Internal Broker do not use a separate file for the direct messaging configuration (previously in the *activemq*-external*.xml file), everything is configured in one *broker.xml* file.

There are multiple *activemq-*.xml* files in ECP Endpoint, so for the migration use only the files which were used by ECP Endpoint.

> */jms/internal-ssl-jks/ if the optional Hardware Security Module (HSM) is not used and internalBroker.useAuthentication is set to true. This was a default configuration.*

> */jms/internal-ssl-hsm/ and /jms/external-ssl-hsm/ if HSM is used*

> */jms/ if optional HSM is not used and internalBroker.useAuthentication is set to false*

File was then selected to match the configured ECP application profile:

> *activemq-nonha(-external).xml when ecp-nonha profile is used. This was a default configuration.*

> *activemq-ha(-external).xml when ecp-ha profile is used*

> *activemq-inmemory(-external).xml when ecp-inmemory profile is used*

In a default configuration, the following file was used:

> */etc/ecp-endpoint/jms/internal-ssl-jks/activemq-nonha.xml*

Compare the matching elements in *activemq-*.xml* and *broker.xml*, and make changes in *broker.xml* if applicable:

> ***transportConnector** element in activemq-*.xml matches **acceptor** element in broker.xml*

- *In the connection string, compare a port number and connection parameters (for example, TLS protocol and ciphers)*
- *When direct messaging was used, create a new acceptor element with a different name and port number. This will be used for the direct messaging connections.*

> ***memoryUsage** element in activemq-*.xml matches **global-max-size** element in broker.xml*

> ***storeUsage** element in activemq-*.xml matches **max-disk-usage** element in broker.xml*

> ***destinationPolicy** element in activemq-*.xml matches **address-settings** element in broker.xml*

ECP Endpoint contains example *broker.xml* files in */etc/ecp-endpoint/jms/ssl-hsm* and *ssl-jks* folders. These files are not used by ECP Endpoint, they are meant only as a support for the Internal Broker *broker.xml* configuration. For example, they show how to configure the direct messaging, or queue memory limits.

If you require more details about the migration process and *broker.xml* configuration, please refer to the official Artemis Migration Guide and Configuration Reference.

Once the configuration migration is complete, all *activemq-*.xml* files can be removed.

5.2.5.2. ECP Endpoint configuration

Please refer to the ECP Installation Guide, chapter *Internal Broker configuration* on how to configure ECP Endpoint to be able to connect to Internal Broker.

5.2.6. Update configuration

5.2.6.1. When upgrading from ECP 4.12.0 or lower

When upgrading from ECP 4.12.0 or lower, the following changes are required.

> ☐ *Open in /etc/ecp-endpoint/**ecp.properties**:*

- *Add this property:*
  - *ecp.csrf.secret*
- *Its value is used as a secret used in a creation of the CSRF token hash for GUI. It should contain random characters, like a password.*
- *These properties are no longer required, so they can be removed:*
  - *spring.activemq.user*
  - *spring.activemq.password*

> ☐ *Open /usr/share/ecp-endpoint/conf/**server.xml**, and find HTTPS connector element*

> ☐ *Add the new HTTPS connector element:*

```
<Connector port="8443" protocol="eu.entsoe.ecp.connector.protocol.EcpHttp11NioProtocol"
sslImplementationName="org.apache.tomcat.util.net.jsse.JSSEImplementation"
maxThreads="150" SSLEnabled="true" maxParameterCount="1000" >
    <UpgradeProtocol className="org.apache.coyote.http2.Http2Protocol" />

    <SSLHostConfig protocols="TLSv1.3"
ciphers="TLS_AES_256_GCM_SHA384,TLS_CHACHA20_POLY1305_SHA256,TLS_AES_128_GCM_SHA256" >
        <Certificate certificateKeystoreFile="/var/lib/ecp-endpoint/authKeystore.jks"
certificateKeystorePassword="<keystore password>" type="RSA" />
    </SSLHostConfig>
</Connector>
```

> ☐ *Copy a value from the old HTTPS connector attribute and pass it to a matching attribute in the new HTTPS connector:*

- *keystorePass to certificateKeystorePassword*
- *any other attribute where a custom change was previously made (for example ciphers)*

> ☐ *Remove the old HTTPS connector element*

Keycloak authentication is no longer supported, so the Keycloak properties can be removed when the Keycloak authentication was previously enabled:

> *Open /etc/ecp-endpoint/**ecp.properties**:*
  - *Remove all properties starting with "keycloak."*
  - *Remove "ecp-sso" profile from the spring.profiles.active*

Please see the *How to configure Keycloak integration with LDAP* chapter in the *ECP Administration Guide* to make a switch to the LDAP authentication.

Optionally, you can remove the *ecp.endpoint.jms.directory* property from the following files:

> ☐ */usr/share/ecp-endpoint/bin/**setenv.sh***

> ☐ */etc/systemd/system/ecp-endpoint.service.d/**env.conf***

The property will be removed automatically during the upgrade if the file was not changed from a default configuration. Otherwise, a file with ***.rpmnew*** will be created and it well be necessary to remove the property manually (or merge previous changes with a file content from the ***.rpmnew***).

As of the Prometheus metrics, the list of previously used ActiveMQ metrics with their Artemis equivalent can be found in the *Artemis Prometheus metrics* chapter.

When Artemis Internal Broker does not use a default keystore, its public certificate must be imported to the ECP Endpoint keystore referenced in the *internalBroker.keystore.location* property. You can find a location of the Artemis keystore in the *keyStorePath* attribute of the acceptor URI in the *broker.xml* file. By default, it is in */opt/eccosp-artemis/etc/keystore.jks*.

Alternatively, it is possible to configure Artemis Internal Broker to use the ECP Endpoint keystore from the *internalBroker.keystore.location* property instead.

5.2.6.2. When upgrading from ECP 4.11.0 or lower

When upgrading from ECP 4.11.0 or lower, the following changes are required.

> ☐ *To enable health endpoint metrics, add these lines in /etc/ecp-endpoint/**ecp.properties**:*
  - *To enable the metrics:*
    - *management.endpoint.health.show-details=ALWAYS*

> *When the LDAP authentication is enabled*

- *The Web Services requests are no longer authenticated by using the users from the **users.properties** file.*
- ☐ *It is necessary to create new users in the LDAP server, or to change the Web Services requests to use users which already exists in the LDAP server.*

5.2.6.3. When upgrading from ECP 4.9.0 or lower

When upgrading from ECP 4.9.0 or lower, the following changes are required.

Verify that there is no *file:* prefix in *-Dspring.config.additional-location* property in the following files:

> ☐ */usr/share/ecp-endpoint/bin/**setenv.sh***

> ☐ */etc/systemd/system/ecp-endpoint.service.d/**env.conf***

The property should look like this:

> *-Dspring.config.additional-location=/etc/ecp-endpoint/ecp.properties,/etc/ecp-endpoint/ecp-users.properties*

The *file:* prefix will be removed automatically during the upgrade if a file was not changed from a default configuration. Otherwise, a file with ***.rpmnew*** will be created and it will be necessary to remove the prefix manually (or merge previous changes with a file content from ***.rpmnew***).

5.2.6.4. MS SQL database

MS SQL database connector is now forcing TLS encryption by default. If you do not use TLS encryption, add *encrypt=false* parameter to the connection string in *ecp.db.url.*

5.2.7. Redeploy plugins

Skip this chapter in case no custom Receive, Sent or Archive Handler was deployed.

ECP plugins (send, receive, archive handlers) must be redeployed after the upgrade. Unzip ***ECP_MODULE.war*** located in *<endpoint install path>/webapps/*. Then copy plugins jar files to *<endpoint install path>/webapps/ECP_MODULE/WEB-INF/lib/*.

5.2.8. Remove Java 11 (only when upgrading from ECP 4.10.1 or lower)

If you have upgraded Java 17 from Java 11, then uninstall Java 11 before starting the application.

! If you are running other ECP/EDX applications on the same server, upgrade them first before uninstalling Java 11. !

> ☐ *yum erase java-11-openjdk.x86_64*

> ☐ *yum erase java-11-openjdk-headless.x86_64*

5.2.9. Start application

After ECP endpoint is configured, it can be started. ECP endpoint is registered as a Linux service. The following command is available:

> *systemctl start ecp-endpoint.service*

If not installed as a service:

> Start application using bash script *"<install_path>/tomcat/bin/startup.sh*"

5.2.10. Installation verification

Check the status of the ecp-endpoint service:

> *systemctl status ecp-endpoint.service*

- The service should be running

Display application log (**ecp-endpoint.log**) output:

> *tail <log_path>/ecp.log*

Display messages from a selected file:

> *tail <log_path>/catalina.out*

- A similar line should appear in the **ecp.log** file: *Started Endpoint in 25.001 seconds (JVM running for 31.924)*

Display the web interface of ECP

> Open URL "*https://<endpoint-ip>:8443/ECP_MODULE*" in your web browser (replace the <endpoint-ip> with the correct IP address of your endpoint server)

- GUI should appear

5.2.10.1.Log files

ECP endpoint log files are available in the */var/log/ecp-endpoint* folder.

5.2.10.2. GUI

It should be possible to open GUI in the web browser. The web browser refresh (Ctrl+Shift+R) or clearing of the web browser cache should be made to make sure that the new version features and changes are loaded.

Everything should be working and configured as it was before the upgrade (except the new version features and changes). Dashboard should be opened after logging into application.

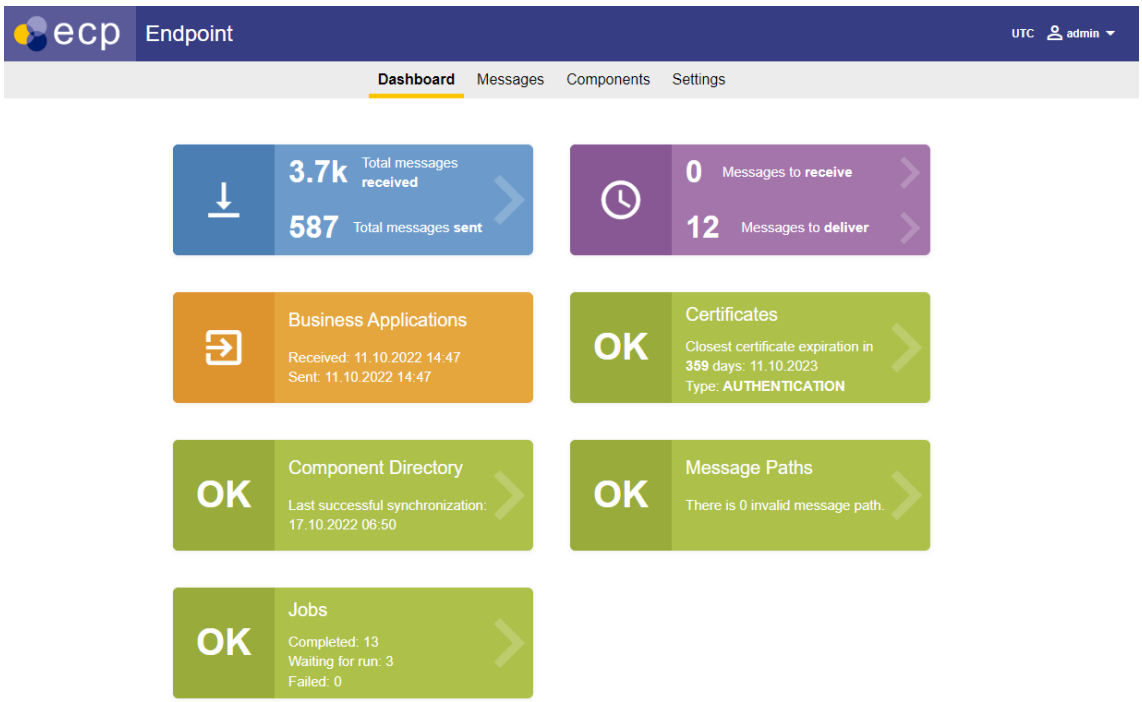

Figure 6: GUI - Verification

5.2.10.3. Apply functional smoke test if needed

Please refer to the *ECP Installation Guide*, chapter *ECP Endpoint Registration and Functional Smoke Test* to apply the test.

5.2.11. Upgrade Notes

Log record time zone is set to UTC, therefore there will be overlap of times in log files.

Default GUI time zone is changed to UTC. Times of old messages are not changed to the new time zone, therefore there can be overlap of times of old and new messages. Time zone of GUI can be changed from GUI Settings page.

Due to a change of the messages indexes in the database, the first start-up after the upgrade may take a longer time when upgrading from ECP 4.14.0 or lower versions. It should take between 20 seconds to 2 minutes, but it depends on number of messages in the database. Reducing the number of messages in the database before the upgrade will help to reduce the start-up time.

ActiveMQ database tables are no longer used by ECP Endpoint, so these tables can be removed from the database:

> *ACTIVEMQ_MSGS*

> *ACTIVEMQ_ACKS*

> *ACTIVEMQ_LOCK*

The pattern of timestamps in the logs has been changed to comply with ISO8601. An example of the current pattern of a timestamp:

> *2024-05-20T14:30Z*

When LDAP authentication is used, the default authentication mode has been changed from the password comparison authentication to the bind authentication. Therefore, it is necessary to verify that the LDAP server supports this authentication mode. There is a new *ecp.ldap.useServerAuthentication* property to configure the authentication mode. This change does not affect the Active Directory authentication.

5.3. New configuration properties

Following configuration properties were added since the last ECP version:

Property	Description	Default value
ecp.disabledAuthenticationActuatorEndpoints	By default, an access to all actuator endpoints exposed on */ECP_MODULE/actuator/** URL is protected by the user authentication. It is possible to disable the authentication for individual endpoints by adding the endpoint name to the comma-separated list in this property.	N/A
ecp.ldap.roleSearchMatching	The filter used in the groups search, with the filter argument {0} being the full DN of the user.	uniqueMember={0}
ecp.ldap.roleSearchSubtree	If set to true, a subtree scope search of groups will be performed, instead of the single-level search.	false

ecp.ldap.rolePrefix	Sets the prefix which will be prepended to the role values loaded from the LDAP directory.	Empty string
ecp.ldap.useServerAuthentication	When set to true, the bind authentication is used, which means that the authentication is performed by LDAP server. Otherwise, the password comparison authentication is used, where LDAP server sends a stored password of the matched user to the client, where the authentication is performed. This mode should always be used with LDAPS, or for testing purposes only.	true
ecp.ldap.searchFilter	The LDAP filter string for Active Directory to search for the user being authenticated. Occurrences of {0} are replaced with the username@domain. Occurrences of {1} are replaced with the username. When AD uses *sAMAccountName* instead of *userPrincipalName*, use this string: *(&(objectClass=user)(sAMAccountName={1}))*	(&(objectClass=user)(userPrincipalName={0}))

5.4. New metrics

No new Prometheus metrics were added since the last ECP version.

The list of previously used ActiveMQ metrics with their Artemis equivalent can be found in the *Artemis Prometheus metrics.*

5.5. Artemis Prometheus metrics

Since the ActiveMQ metrics are no longer available, the following Artemis metrics can be used as an equivalent for some metrics. The Artemis metrics are available only in a connected Internal Broker, not in ECP Endpoint.

ActiveMQ metric	Artemis metric
ecp_internal_broker_total_connections_count	artemis_total_connection_count
ecp_internal_broker_total_producer_count	-
ecp_internal_broker_memory_limit	-
ecp_internal_broker_memory_limit_percent_usage	artemis_address_memory_usage_percentage
ecp_internal_broker_store_limit	-
ecp_internal_broker_store_percent_usage	artemis_disk_store_usage
ecp_internal_broker_queue_average_enqueue_time	-
ecp_internal_broker_queue_consumer_count	artemis_consumer_count
ecp_internal_broker_queue_dequeue_count	artemis_routed_message_count
ecp_internal_broker_queue_enqueue_count	artemis_messages_added
ecp_internal_broker_queue_expired_count	artemis_messages_expired
ecp_internal_broker_queue_in_flight_count	artemis_delivering_message_count
ecp_internal_broker_queue_memory_limit	-
ecp_internal_broker_queue_memory_percent_usage	artemis_address_size
ecp_internal_broker_queue_queue_size	artemis_message_count

6. Internal Broker Upgrade

6.1. Upgrade on Windows

6.1.1. Variables used in the guide

Variables used throughout this upgrade guide are using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<install_path>	Path pointing to the installation directory

6.1.2. Stop application

6.1.2.1. Using Command Line

Stop service using

> *SC stop eccosp-artemis*

6.1.2.2. Using GUI

> *Start > Run > "services.msc"*

> *Find service with name like "eccosp-artemis"*

> *Stop service by button on the top left*

6.1.3. Java

Check the current Java version using the following command:

> ☐ *java -version*

The expected output is *openjdk version "17.0.14"* or newer versions. If the version is below *17.0.14* or the 0.14-update is older than 6 months, the update is required.

6.1.4. Install new version of application

For installation of the new version .jar go through following steps.

GUI INSTALLATION

1. ☐ *Double-click the <eccosp-artemis-jar> file. If Java was registered to execute jar files, an installer window should open. Skip to step 5. If an installer window does not open, follow steps 2-4:*
2. ☐ *Open command line with the administrator rights:*
   > *Start > type "cmd" > right click > run as administrator*
3. ☐ *Change directory to location of **eccosp-artemis.jar**, e.g.*
   > *cd c:\eccosp-artemis-installer*
4. ☐ *Run the **eccosp-artemis.jar** file using*
   > *java -jar <eccosp-artemis-jar>*
5. *Internal Broker installer window is opened.*
   > ☐ *Select installation type – Upgrade an existing installation*
   > ☐ *Select target installation path (the target directory must exist)*

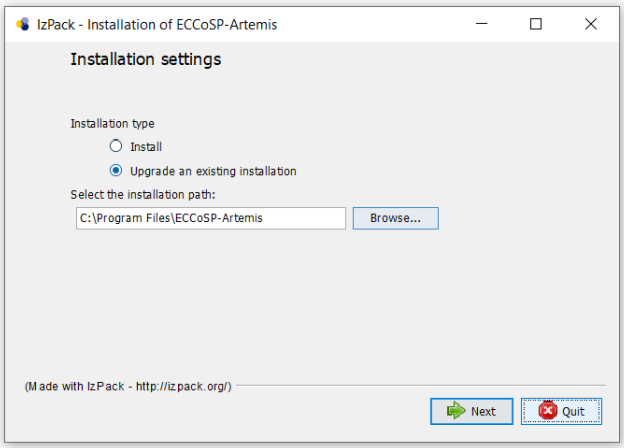

Figure 7: Installation 1

   > ☐ *Click Next*
   > ☐ *Installation is processed*
   > ☐ *Click Next*
   > ☐ *If desired, an automatic installation script can be generated. It can be used in the future to install the application using the same options.*

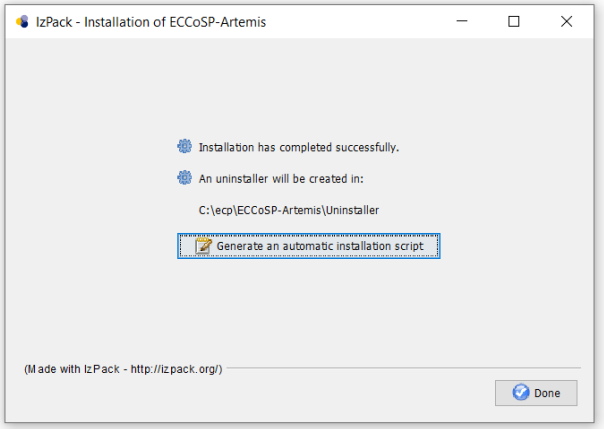

Figure 8: Installation 2

> ☐ *Click Done*

COMMAND LINE INSTALLATION

> ☐ *Open command line with the administrator rights:*

- *Start > type "cmd" > right click > run as administrator*

> ☐ *Change directory to location of <eccosp-artemis-jar>, e.g.*

- *cd c:\eccosp-artemis-installer*

> ☐ *Run the eccosp-artemis.jar file using*

- *java -jar < eccosp-artemis-jar> -console*

> ☐ *Follow the console installation wizard*

6.1.5. Installation verification

The application should be installed in the <install_path> folder. This installation folder should contain Artemis core files, broker instance folder, and an uninstaller.

6.1.5.1. Service status

The status of the service, if installed, can be checked either via command line:

- *SC query eccosp-artemis*

or in the Windows Services tool.

6.1.5.2. Log files

Check the application log files for more information about its status. The default location is *<install_path>\broker\log*.

6.1.5.3. GUI

GUI of Artemis Broker is started on port 8161.

6.1.6. Upgrade Notes

The pattern of timestamps in the logs has been changed to comply with ISO8601. An example of the current pattern of a timestamp:

> *2024-05-20T14:30Z*

6.2. Upgrade on Linux

6.2.1. Stop application

The following command is available:

> *systemctl stop eccosp-artemis*

6.2.2. Java

Check the current Java version using the following command:

> ☐ *java -version*

The expected output is *openjdk version "17.0.14"* or newer versions. If the version is below *17.0.14* or the 0.14-update is older than 6 months, the update is required.

Run the following command for the Java update:

> ☐ *yum update java-17-openjdk*

6.2.3. Internal Broker component upgrade

Change directory to the folder where the Internal Broker rpm is located and invoke rpm upgrade.

> ☐ *rpm -U <eccosp-artemis-rpm>*

It keeps original configuration files and it creates new configuration files with a ***.rpmnew*** suffix. File ***.rpmnew*** is not created in case that original configuration was not modified.

By default, Internal Broker is set up to automatically start after server start. The automatic start can be disabled by the following command:

> ☐ *systemctl disable eccosp-artemis*

Following folders are updated:

Folder	Description
/usr/share/eccosp-artemis	Folders for Artemis core files
/opt/eccosp-artemis/lib	Folders for Artemis broker instance libraries
/opt/eccosp-artemis/etc	Artemis broker instance configuration files

6.2.4. Start application

The following command is available:

> *systemctl start eccosp-artemis*

6.2.5. Installation verification

6.2.5.1. Broker status

The status of the broker can be checked via command line:

> *systemctl status eccosp-artemis*

6.2.5.2. Log files

Check the application log files for more information about its status. The location is */opt/eccosp-artemis/log*.

6.2.5.3. Artemis GUI

GUI of Artemis Broker is started on port 8161.

6.2.6. Upgrade Notes

The pattern of timestamps in the logs has been changed to comply with ISO8601. An example of the current pattern of a timestamp:

> *2024-05-20T14:30Z*

6.3. New configuration properties

No new configuration properties were added since the last ECP version.

6.4. New metrics

No new Prometheus metrics were added since the last ECP version.

7. ECP Component Directory Upgrade

If you are upgrading Component Directory in the High Availability installation, ECP does not support an upgrade with a zero downtime – it is not possible for the running nodes to have a different ECP versions. This procedure is recommended for an upgrade with a minimal outage:

1. *Stop one of the Component Directory nodes*
2. *Upgrade the stopped node*
3. *Stop the other nodes*
4. *Start the upgraded node*
5. *Upgrade all remaining nodes and start them*

7.1.1. Variables used in the guide

Variables used throughout this upgrade guide are using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<install_path>	Path pointing to the installation directory

7.1.2. Stop application

The following command is available:

> *systemctl stop ecp-directory.service*

If not installed as a service:

> *Stop application using bash script "<install_path>/tomcat/bin/shutdown.sh"*

7.1.3. Java

Check the current Java version using the following command:

> ☐ *java -version*

The expected output is *openjdk version "17.0.14" or newer versions. If the version is below 17.0.14* or the 0.14-update is older than 6 months, the update is required.

7.1.3.1. Upgrade to Java 17 (only when upgrading from ECP 4.10.1 or lower)

An upgrade of Java is necessary prior installation of the new application version.

If your Java version is 11, run the following command to install Java 17:

> ☐ *yum install java-17-openjdk*

The minimal Java version is 17.0.14. If your Java version is lower, run the following command for Java update:

> ☐ *yum update java-17-openjdk*

After Java installation, configure Java properties:

> ☐ *Open the security configuration file, e.g. on path: /usr/lib/jvm/java-17-openjdk-17.0.14.0.7-2.0.1.el9.x86_64/conf/security/java.security*

> ☐ *Change the line securerandom.source=file:/dev/random to securerandom.source=file:/dev/urandom*

> ☐ *Save the file*

! Do not uninstall Java 11 at this moment, First upgrade ECP Component Directory – follow the steps below. !

7.1.4. ECP Component Directory application upgrade

Change the directory to the folder where the ECP rpm package is located and then invoke rpm upgrade command.

> ☐ *rpm -U <ecp-rpm>*

It keeps original configuration files and it creates new configuration files with a ***.rpmnew*** suffix. File ***.rpmnew*** is not created in case that original configuration was not modified.

By default, ECP Component Directory is set up to automatically start after server start. The automatic start can be disabled by the following command:

> ☐ *systemctl disable ecp-directory*

Following folders are updated:

Folder	Description
/usr/share/ecp-directory	Application binaries folder. This is the default location of <install_path>
/var/lib/ecp-directory	Application data folder

/etc/ecp-directory	ECP component directory configuration files
/etc/systemd/system/ecp-directory.service.d/	Default configuration of ECP component directory. Used to set JVM configuration and location of configuration files. (file /etc/systemd/system/ecp-directory.service.d/env.conf)

7.1.5. Update configuration

7.1.5.1. When upgrading from ECP 4.12.0 or lower

When upgrading from ECP 4.12.0 or lower, the following changes are required.

> ☐ *Open in /etc/ecp-directory/**ecp-directory.properties**:*

- *Add this property:*
  - *ecp.csrf.secret*
- *Its value is used as a secret used in a creation of the CSRF token hash for GUI. It should contain random characters, like a password.*

> ☐ *Open /usr/share/ecp-directory/conf/**server.xml**, and find HTTPS connector element*

> ☐ *Add the new HTTPS connector element:*

```
<Connector port="8443"
protocol="eu.entsoe.ecp.connector.protocol.EcpDirectoryHttp11NioProtocol"
sslImplementationName="org.apache.tomcat.util.net.jsse.JSSEImplementation"
maxThreads="150" SSLEnabled="true" maxParameterCount="1000" >
    <UpgradeProtocol className="org.apache.coyote.http2.Http2Protocol" />

    <SSLHostConfig protocols="TLSv1.3" truststoreFile="/var/lib/ecp-
directory/authKeystore.jks" truststorePassword="<truststore password>"
certificateVerification="optional"
ciphers="TLS_AES_256_GCM_SHA384,TLS_CHACHA20_POLY1305_SHA256,TLS_AES_12
8_GCM_SHA256" >
        <Certificate certificateKeystoreFile="/var/lib/ecp-directory/authKeystore.jks"
certificateKeystorePassword="<keystore password>" certificateKeyAlias="ecp_module_auth"
type="RSA" />
    </SSLHostConfig>
 </Connector>
```

> ☐ *Copy a value from the old HTTPS connector attribute and pass it to a matching attribute in the new HTTPS connector:*
  - *truststorePass to truststorePassword*
  - *keystorePass to certificateKeystorePassword*
  - *any other attribute where a custom change was previously made (for example ciphers)*

> ☐ *Remove the old HTTPS connector element*

Keycloak authentication is no longer supported, so the Keycloak properties can be removed when the Keycloak authentication was previously enabled:

> *Open /etc/ecp-directory/**ecp-directory.properties**:*
  - *Remove all properties starting with "keycloak."*
  - *Remove "ecp-sso" profile from the spring.profiles.active*

Please see the *How to configure Keycloak integration with LDAP* chapter in the *ECP Administration Guide* to make a switch to the LDAP authentication.

7.1.5.2. When upgrading from ECP 4.11.0 or lower

When upgrading from ECP 4.11.0 or lower, the following changes are required.

> ☐ *To enable health endpoint metrics, add this line in /etc/ecp-directory/**ecp-directory.properties**:*
  - *management.endpoint.health.show-details=ALWAYS*

7.1.5.3. When upgrading from ECP 4.9.0 or lower

When upgrading from ECP 4.9.0 or lower, the following changes are required.

Verify that there is no *file:* prefix in *-Dspring.config.additional-location* property in the following files:

> ☐ */usr/share/ecp-directory/bin/**setenv.sh***

> ☐ */etc/systemd/system/ecp-directory.service.d/**env.conf***

The property should look like this:

> *-Dspring.config.additional-location=/etc/ecp-directory/ecp.properties,/etc/ecp-directory/ecp-users.properties*

The *file:* prefix will be removed automatically during the upgrade if a file was not changed from a default configuration. Otherwise, a file with ***.rpmnew*** will be created and it will be necessary to remove the prefix manually (or merge previous changes with a file content from ***.rpmnew***).

7.1.5.4. MS SQL database

MS SQL database connector is now forcing TLS encryption by default. If you do not use TLS encryption, add *encrypt=false* parameter to the connection string in *ecp.db.url.*

7.1.6. Remove Java 11 (only when upgrading from ECP 4.10.1 or lower)

If you have upgraded Java 17 from Java 11, uninstall Java 11 before starting the application.

! If you are running other ECP/EDX applications on the same server, upgrade them first before uninstalling Java 11. !

> ☐ *yum erase java-11-openjdk.x86_64*

> ☐ *yum erase java-11-openjdk-headless.x86_64*

7.1.7. Start application

After ECP Component Directory is configured, it can be started. ECP Component Directory is registered as a Linux service. The following command is available:

> *systemctl start ecp-directory.service*

If not installed as a service:

> *Start application using bash script "<install_path>/tomcat/bin/startup.sh"*

7.1.8. Installation verification

Check the status of the ecp-directory service:

> *systemctl status ecp-directory.service*

- *The service should be running.*

Display the application log (**ecp.log**) output:

> *tail <log_path>/ecp.log*

Display messages from a selected file:

> *tail <log_path>/catalina.out*

- *A similar line should appear in the **ecp.log** file: Started Component Directory in 17.785 seconds (JVM running for 25.084)*

7.1.8.1. GUI

It should be possible to open GUI in the web browser. The web browser refresh (Ctrl+Shift+R) or clearing of the web browser cache should be made to make sure that the new version features and changes are loaded.

Everything should be working and configurated as it was before the upgrade (except the new version features and changes). Dashboard should be opened after logging into application.

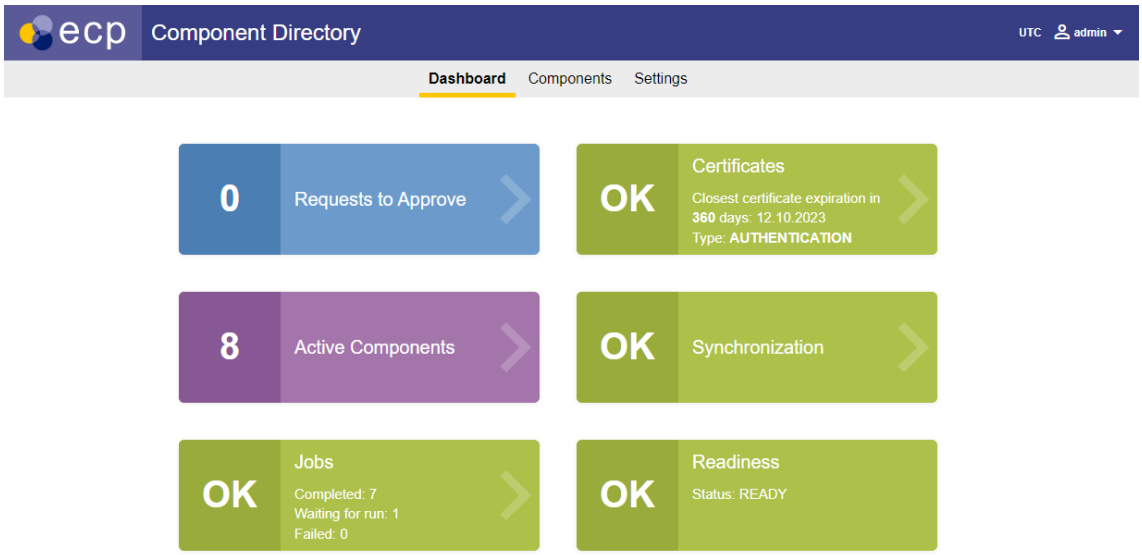

Figure 9: GUI - Verification

7.1.9. Upgrade Notes

Log record time zone is set to UTC, therefore there will be overlap of times in log files.

Default GUI time zone is changed to UTC. Time zone of GUI can be changed from GUI Settings page.

The *ecp.pushes.directory.pushedTtlThreshold* property was removed. The CD TTL is now updated with every successful synchronization.

The pattern of timestamps in the logs has been changed to comply with ISO8601. An example of the current pattern of a timestamp:

> *2024-05-20T14:30Z*

When LDAP authentication is used, the default authentication mode has been changed from the password comparison authentication to the bind authentication. Therefore, it is necessary to verify that the LDAP server supports this authentication mode. There is a new *ecp.ldap.useServerAuthentication* property to configure the authentication mode. This change does not affect the Active Directory authentication.

7.2. New configuration properties

Following configuration properties were added since the last ECP version:

Property	Description	Default value
ecp.disabledAuthenticationActuatorEndpoints	By default, an access to all actuator endpoints exposed on */ECP_MODULE/actuator/** URL is protected by the user authentication. It is possible to disable the authentication for individual endpoints by adding the endpoint name to the comma-separated list in this property.	N/A
ecp.ldap.useServerAuthentication	When set to true, the bind authentication is used, which means that the authentication is performed by LDAP server. Otherwise, the password comparison authentication is used, where LDAP server sends a stored password of the matched user to the client, where the authentication is performed. This mode should always be used with LDAPS, or for testing purposes only.	true
ecp.ldap.searchFilter	The LDAP filter string for Active Directory to search for the user being authenticated. Occurrences of {0} are replaced with the username@domain. Occurrences of {1} are replaced with the username. When AD uses *sAMAccountName* instead of *userPrincipalName*, use this string: *(&(objectClass=user)(sAMAccountName={1}))*	(&(objectClass=user)(userPrincipalName={0}))

7.3. New metrics

No new Prometheus metrics were added since the last ECP version.

8. ECP Broker Upgrade

8.1.1. Stop application

The following command is available:

> *systemctl stop ecp-broker*

8.1.2. Java

Check the current Java version using the following command:

> ☐ *java -version*

The expected output is *openjdk version "17.0.14"* or newer versions. If the version is below *17.0.14* or the 0.14-update is older than 6 months, the update is required.

8.1.2.1. Upgrade to Java 17 (only when upgrading from ECP 4.10.1 or lower)

An upgrade of Java is necessary prior installation of the new application version.

If your Java version is 11, run the following command to install Java 17:

> ☐ *yum install java-17-openjdk*

The minimal Java version is 17.0.14. If your Java version is lower, run the following command for Java update:

> ☐ *yum update java-17-openjdk*

After Java installation, configure Java properties:

> ☐ *Open the security configuration file, e.g. on path: /usr/lib/jvm/java-17-openjdk-17.0.14.0.7-2.0.1.el9.x86_64/conf/security/java.security*

> ☐ *Change the line securerandom.source=file:/dev/random to securerandom.source=file:/dev/urandom*

> ☐ *Save the file*

! Do not uninstall Java 11 at this moment, First upgrade ECP Broker – follow the steps below. !

8.1.3. ECP Broker upgrade (when upgrading from ECP 4.14.0 or higher)

Change directory to the folder where the ECP Broker RPM is located and invoke rpm upgrade.

> ☐ *rpm -U <ecp-rpm>*

It keeps original configuration files and it creates new configuration files with a ***.rpmnew*** suffix. File ***.rpmnew*** is not created in case that original configuration was not modified.

By default, ECP Broker is set up to automatically start after server start. The automatic start can be disabled by the following command:

> ☐ *systemctl disable ecp-broker*

Following folders are updated:

Folder	Description
/opt/ecp-broker/artemis	Folders for Artemis core files
/opt/ecp-broker/broker/lib	Folders for Artemis broker instance libraries
/opt/ecp-broker/broker/etc	Artemis broker instance configuration files
/opt/ecp-broker/reg	ECP Broker registration tool

8.1.4. ECP Broker upgrade (when upgrading from ECP 4.12.0 or lower)

Due to a major change of switching from ActiveMQ to Artemis broker, ECP Broker rpm does not support the upgrade option in rpm. It is necessary to uninstall the ECP Broker, make a backup, install a new version with ECP Broker rpm, and restore the configuration from the backup. Please note, that all pending messages will be lost during the upgrade, therefore it is recommended to let the ECP Broker process all pending messages prior the upgrade.

In the next release, the rpm upgrade option will be supported again, when upgrading from ECP version with Artemis.

Make a backup of the following folders:

> ☐ */opt/ecp-broker/activemq/conf*

> ☐ */opt/ecp-broker/cd*

> ☐ */opt/ecp-broker/config*

Uninstall the current ECP Broker installation.

> ☐ *rpm -e ecp-broker*

Install the new ECP Broker version. Please refer to the ECP Installation Guide, chapter *ECP Broker Installation* on how to install the ECP Broker.

By default, ECP Broker is set up to automatically start after server start. The automatic start can be disabled by the following command:

> ☐ *systemctl disable ecp-broker*

Installer automatically creates the following folder structure:

Folder	Description
/opt/ecp-broker/artemis	Folders for Artemis core files
/opt/ecp-broker/broker	Folders for Artemis broker instance
/opt/ecp-broker/broker/etc	Artemis broker instance configuration files
/opt/ecp-broker/broker/data	Artemis broker instance data
/opt/ecp-broker/cd	ECP Component Directory content storage
/opt/ecp-broker/config	ECP Broker configuration files
/opt/ecp-broker/reg	ECP Broker registration tool

Copy backup files to these folders, overwriting installed files.

> ☐ */opt/ecp-broker/cd*

> ☐ */opt/ecp-broker/config*

ActiveMQ broker configuration in *activemq.xml* file previously located in /opt/*ecp-broker/activemq/conf* is no longer used. Instead, the Artemis uses the configuration in *broker.xml* file located in */opt/ecp-broker/broker/etc/*.

When custom changes were previously made in *activemq.xml*, it is necessary to migrate these changes to *broker.xml*.

Compare the matching elements in *activemq.xml* and *broker.xml*, and make changes in *broker.xml* if applicable:

> ***transportConnector*** *element in activemq.xml matches* ***acceptor*** *element in broker.xml*
>
> - *In the connection string, compare a port number and connection parameters (for example, TLS protocol and ciphers)*

> ***memoryUsage*** *element in activemq.xml matches* ***global-max-size*** *element in broker.xml*

> ***storeUsage*** *element in activemq.xml matches* ***max-disk-usage*** *element in broker.xml*

> ***destinationPolicy*** *element in activemq.xml matches* ***address-settings*** *element in broker.xml*

If you require more details about the migration process and *broker.xml* configuration, please refer to the official Artemis Migration Guide and Configuration Reference.

8.1.5. Update configuration

If there are changes in the **broker.properties** file, the configuration has to be pushed into ECP Component Directory. Change the current directory to */opt/ecp-broker/reg* where the broker-registration-tool.jar is located and use the following command to push the configuration:

> ☐ *java -D"spring.config.location=/opt/ecp-broker/config/broker.properties" -jar broker-registration-tool.jar -m PUSH_CONFIGURATION*

Or using socks proxy configuration:

> ☐ *java -D"spring.config.location=/opt/ecp-broker/config/broker.properties" -D"socksProxyHost=<ProxyIP>" -D"socksProxyPort=<ProxyPort>" -jar broker-registration-tool.jar -m PUSH_CONFIGURATION*

8.1.6. Remove Java 11 (only when upgrading from ECP 4.10.1 or lower)

If you have upgraded Java 17 from Java 11, uninstall Java 11 before starting the application.

> ☐ *yum erase java-11-openjdk.x86_64*

> ☐ *yum erase java-11-openjdk-headless.x86_64*

8.1.7. Start application

Before starting the broker make sure that you have successfully pushed the configuration. The following command is available:

> *systemctl start ecp-broker*

8.1.8. Installation verification

The application should be installed in the /opt/ecp-broker folder.

8.1.8.1. Broker status

The status of the broker can be checked via command line:

> *systemctl status ecp-broker*

8.1.8.2. Log files

Check the application log files for more information about its status. The location is */opt/ecp-broker/broker/log*.

8.1.8.3. Artemis GUI

GUI of Artemis Broker is started on port 8161. By default, the access is limited for opening it on the server where the Artemis is installed.

For allowing connection to GUI from other locations it is necessary to change the configuration file */opt/ecp-broker/broker/etc/jolokia-access.xml.*

> *Add <allow-origin> element with an URL pattern to match URLs from where the access will be allowed*

> *Multiple <allow-origin> elements can be defined*

> *URL pattern supports a wildcard character: **

8.1.9. Upgrade Notes

The pattern of timestamps in the logs has been changed to comply with ISO8601. An example of the current pattern of a timestamp:

> *2024-05-20T14:30Z*

8.2. New configuration properties

No new configuration properties were added since the last ECP version.

8.3. New metrics

No new Prometheus metrics were added since the last ECP version.

9. Kubernetes Support

ECP installer packages provide Helm charts and Values files. To learn how to deploy ECP components in the Kubernetes cluster, please see the ECP Kubernetes Guide.

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